Status	Finished
Started	Tuesday, 20 October 2026, 9:27 PM
Completed	Tuesday, 20 October 2026, 9:33 PM
Duration	5 mins 5 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It is slower than linear search for small lists
- b. It can only be applied to large lists
- c. It requires the list to be sorted
- d. It does not work with negative numbers

Question **2**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- a. 90 and 94
- b. 89 and 99
- c. 90 and 99
- d. 89 and 94

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Merge search
- b. Quick search
- c. Bubble search
- d. Linear search

Question **4**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It can be applied only if the list is sorted ✓
- b. It always starts from the beginning of the list
- c. It works on unsorted lists
- d. It compares elements sequentially

Question **5**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n ✓
- b. log n
- c. n-1
- d. n/2

Question **6**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By dividing the list into halves
- b. By using a hash function
- c. By comparing each element in the list sequentially ✓
- d. By sorting the list first

Question **7**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search stops immediately ✓
- b. The search continues until the end of the list
- c. The search starts over from the beginning
- d. The search backtracks to find duplicate elements

Question **8**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 100
- b. 500
- c. 10 □
- d. 1000

Question **9**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Jump search
- b. Binary search
- c. Linear search □
- d. Hashing

Question **10**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search works on unsorted lists
- b. Binary search does not require dividing the list
- c. Binary search can find multiple instances of the target element
- d. Binary search has a lower time complexity on large, sorted lists □

Question **11**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Hash search

b. Both (1) & (3)

c. Linear search

d. Binary search

Question **12**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist

b. The list is sorted

c. The search ends successfully

d. The search continues in the right sublist

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.

b. Linear search requires the list to be sorted.

c. Linear search can be applied to both sorted and unsorted lists.

d. Linear search is more efficient than binary search on large lists.

Question **14**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By dividing the list length by two
- b. By using a hash function
- c. By comparing each element sequentially
- d. By starting from the first element

Question **15**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Hash search
- b. Linear search
- c. Binary search
- d. Sequential search